

▼ NLP

Start coding or [generate](#) with AI.

```
text='HI,HOW ARE YOU'
text_lower=text.lower()
print(text_lower)
```

hi,how are you

```
import re
text='hello This Is Aswitha'
tect_clean=re.sub(r'^a-z',' ', text)
print(tect_clean)
```

hello his s switha

```
import re
text='hello This Is Aswitha'
tect_clean=re.sub(r'[a-z]',' 2' , text)
print(tect_clean)
```

2 2 2 2 2 T 2 2 2 I 2 A 2 2 2 2 2 2

```
import nltk
from nltk.corpus import stopwords
nltk.download('stopwords',quiet=True)
stop_words=set(stopwords.words('english'))
text=" I'm going to college today"
tokens=[]
for word in text.split():
    if word not in stop_words:
        tokens.append(word)
print(tokens)
```

["I'm", 'going', 'college', 'today']

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```
import nltk
nltk.download("punkt_tab",quiet=True)
text="I'm going to college today"
basic=text.split()
print('basic split',basic)
from nltk.tokenize import word_tokenize
smart=word_tokenize(text.lower())
print('nltk tokenize:',smart)
```

basic split ["I'm", 'going', 'to', 'college', 'today']
nltk tokenize: ['i', "m", 'going', 'to', 'college', 'today']

```
import nltk
from nltk.stem import porterStemmer,wordnetLemmatizer
nltk.download('wordnet',quiet=True)
stemmer = porterStemmer()
Lemmatizer=wordnetLemmatizer()
words=['running','studies','better','caring','happily']
for word in words:
    stem=stemmer.stem(word)
    lemma=Lemmatizer.lemmatize(word)
    print(f'{word:12} | {stem} | {lemma}')
```

```

-----
ImportError                                Traceback (most recent call last)
/tmp/ipykernel_873/2318658141.py in <cell line: 0>()
      1 import nltk
----> 2 from nltk.stem import porterStemmer,wordnetLemmatizer
      3 nltk.download('wordnet',quiet=True)
      4 stemmer = porterStemmer()
      5 Lemmatizer=wordnetLemmatizer()

ImportError: cannot import name 'porterStemmer' from 'nltk.stem' (/usr/local/lib/python3.12/dist-packages/nltk/stem/__init__.py)

```

NOTE: If your import is failing due to a missing package, you can manually install dependencies using either `!pip` or `!apt`.

To view examples of installing some common dependencies, click the "Open Examples" button below.

OPEN EXAMPLES

```

import re
import nltk
from nltk.stem import PorterStemmer, WordNetLemmatizer
nltk.download('wordnet',quiet=True)
stemmer=PorterStemmer()
lemmatizer=WordNetLemmatizer()
words=['running','studies','better','caring','happily']
for word in words:
    stemmed_word=stemmer.stem(word)
    lemma=lemmatizer.lemmatize(word,pos='v')
    print(f'{word:12}stem={stemmed_word}lemma={lemma}')

```

```

running    stem=runlemma=run
studies    stem=studilemma=study
better     stem=betterlemma=better
caring     stem=carelemma=care
happily    stem=happililemma=happily

```

```

def preprocess(text):
    text=text.lower()
    text=re.sub(r'^[a-z]', ' ',text)
    tokens=text.split()
    filtered_tokens=[]
    for w in tokens:
        if w not in stop_words:
            filtered_tokens.append(w)
    return filtered_tokens
text="I'm going to college today"
tokens=preprocess(text)
print(tokens)
lemmatized_tokens=[]
for w in tokens:
    lemmatized_tokens.append(
        lemmatizer.lemmatize(w))
print(lemmatized_tokens)
return ' '.join(tokens)
print(preprocess('i am running to the store'))

```

```

['going', 'college', 'today']
['going', 'college', 'today']
File "/tmp/ipykernel_873/1167160840.py", line 18
    return ' '.join(tokens)
    ^
SyntaxError: 'return' outside function

```

```

from sklearn.feature_extraction.text import CountVectorizer
emails=[
    'cricket match cricket match cricket',
    'free win money free win money win',
    'cricket match free win money cricket',
]
cv=CountVectorizer()
x=cv.fit_transform(emails).toarray()
print('vocabulary:',cv.get_feature_names_out())
print('bow matrix:')
print(x)

```

```
vocabulary: ['cricket' 'free' 'match' 'money' 'win']  
bow matrix:  
[[3 0 2 0 0]  
 [0 2 0 2 3]  
 [2 1 1 1 1]]
```

```
from sklearn.feature_extraction.text import TfidfVectorizer  
emails=[  
    'cricket match cricket match cricket',  
    'free win money free win money win',  
    'cricket match free win money cricket',  
]  
tfidf=TfidfVectorizer(sublinear_tf=True,max_df=0.95,min_df=1)  
x_tfidf=tfidf.fit_transform(emails).toarray()  
print('features names:',tfidf.get_feature_names_out())  
print('tf_idf matrix(rounded):')  
import numpy as np  
print(np.round(x_tfidf,2))
```

```
features names: ['cricket' 'free' 'match' 'money' 'win']  
tf_idf matrix(rounded):  
[[0.78 0.    0.63 0.    0.   ]  
 [0.    0.53 0.    0.53 0.66]  
 [0.65 0.38 0.38 0.38 0.38]]
```